

第一次单元测试参考解答

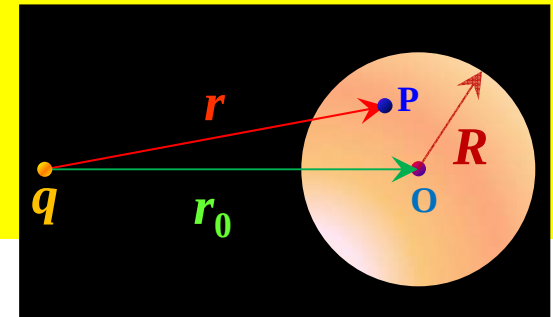
请同学们除了参考自己班级的参考解答，建议其他两个班级的试题也做一下，因为每个班考察的知识点不完全相同。

A1-1. 如图所示,一不带电的**导体球**,若在距球心为 r_0 的处,放一电量为 q 的点电荷,试求:

(1) 球上的感应电荷在球内P点的场强 E_P' 和电势 U_P' ;

(2) 若将球接地,则此时球上的感应电荷在球内P点的 E_P' 和 U_P' ;

(3) 球接地,导体球表面总的感应电荷 q' .



解: (1)金属球不接地时

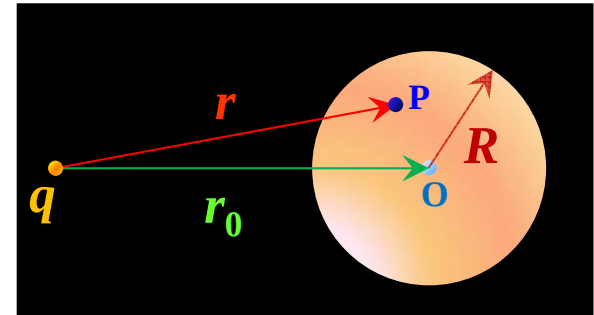
由场强叠加原理,P点的电场强度为点电荷 q 和球面感应电荷单独存在时在该处产生电场强度的矢量和:

$$\vec{E}_P = \vec{E}_{qP} + \vec{E}'_P = \frac{q}{4\pi\epsilon_0 r^2} \hat{r} + \vec{E}'_P$$

根据静电平衡条件,金属导体内部的场强为0.

$$\vec{E}_P = \frac{q}{4\pi\epsilon_0 r^2} \hat{r} + \vec{E}'_P = 0 \quad \text{于是} \quad \vec{E}'_P = -\frac{q}{4\pi\epsilon_0 r^2} \hat{r}$$

由原理叠加原理, P点的电势为点电荷 q 和球面感应电荷单独存在时在该处产生电势的标量和:



$$U_P = U_{qP} + U'_P = \frac{q}{4\pi\epsilon_0 r} + U'_P$$

根据静电平衡条件,金属导体是等势体,故导体球内任一点的电势与球心等势.

$$U_P = U_O = \frac{q}{4\pi\epsilon_0 r_0} + U'_O$$

由于金属球原先不带电,故感应电荷量的代数和为0,即净电荷为0. 于是感应电荷在球心产生的电势为0,即 $U'_O = 0$.

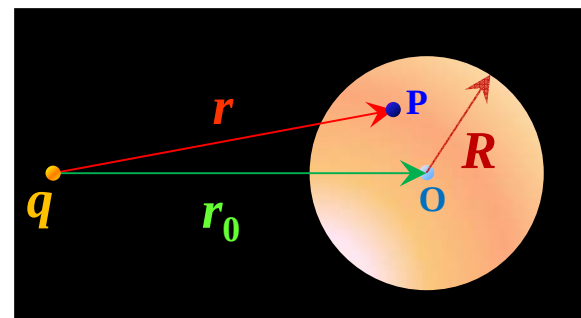
$$U_P = U_O = \frac{q}{4\pi\epsilon_0 r_0}$$

金属球感应电荷在P点产生的电势为:

$$U_P = \frac{q}{4\pi\epsilon_0 r} + U'_P = \frac{q}{4\pi\epsilon_0 r_0} \quad U'_P = \frac{q}{4\pi\epsilon_0 r_0} - \frac{q}{4\pi\epsilon_0 r}$$

(2)金属球接地时,球上的感应电荷在球内P点的 E'_P 和 U'_P

根据静电平衡条件,金属导体是等势体,金属球接地后,故导体球任一点的电势 $U=0$.



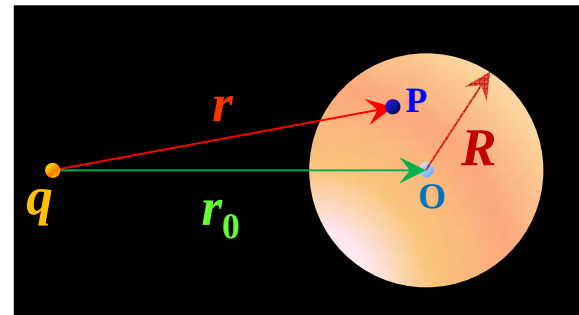
$$U_P = U_{qP} + U'_P = \frac{q}{4\pi\epsilon_0 r} + U'_P = 0 \quad U'_P = -\frac{q}{4\pi\epsilon_0 r}$$

根据静电平衡条件,金属导体内部的场强为0.

$$\vec{E}_P = \frac{q}{4\pi\epsilon_0 r^2} \hat{r} + \vec{E}'_P = 0 \quad \text{于是} \quad \vec{E}'_P = -\frac{q}{4\pi\epsilon_0 r^2} \hat{r}$$

(3) 金属球接地时, 导体球表面总的感应电荷 q' .

根据静电平衡条件, 金属导体是等势体, 金属球接地后, 故导体球任一点的电势 $U=0$, 球心O的电势也为0.



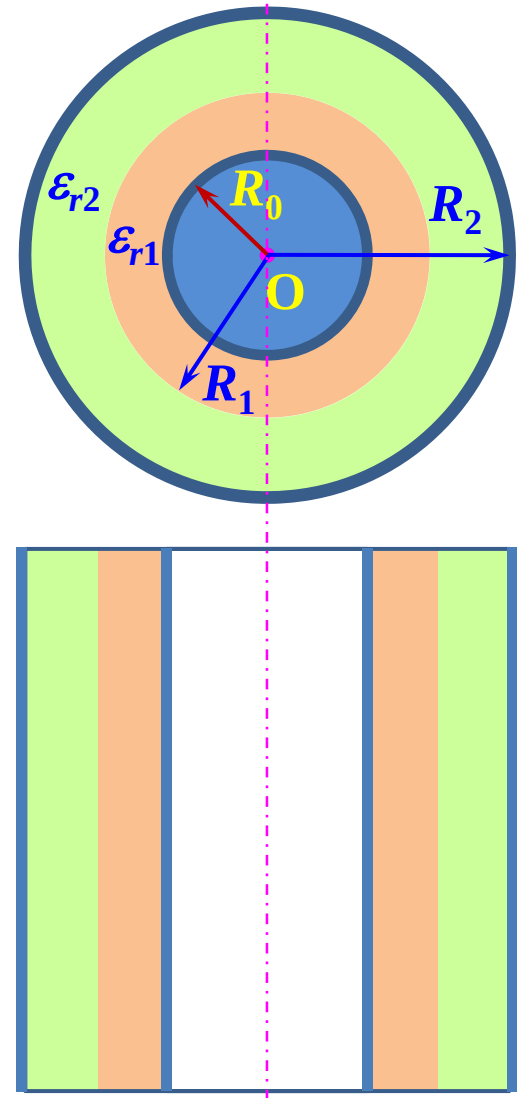
虽然接地后, 导体球表面的感应电荷 q' 在球面上分布是不均匀的, 但感应电荷到O的距离都相等. 于是感应电荷 q' 在球心O产生的电势为:

$$U_{q'O} = \int_{q'} \frac{dq'}{4\pi\epsilon_0 R} = \frac{1}{4\pi\epsilon_0 R} \int_{q'} dq' = \frac{q'}{4\pi\epsilon_0 R}$$

由原理叠加原理, 球心O点的电势为点电荷 q 和球面感应电荷单独存在时在O处产生电势的标量和:

$$U_O = U_{qO} + U_{q'O} = \frac{q}{4\pi\epsilon_0 r_0} + \frac{q'}{4\pi\epsilon_0 R} = 0 \quad q' = -\frac{R}{r_0} q$$

A1-2. 如图所示, 长为 l 的圆柱形电容器, 由半径为 R_0 的圆柱形直导体及共轴的导体外圆筒构成, 其间充满相对介电常量为 ϵ_{r1} 和 ϵ_{r2} 的两层介质. 设电容器带电, 单位长度的电量为 λ_0 , 试求: (1) 介质内的 D 、 E ; (2) 导线和圆筒间的电势差; (3) 介质内的 P 及介质表面的极化电荷面密度 σ' ; (4) 该电容器的电容 C .



解: (1) 用高斯定理可求

$$R_0 < r < R_1 \text{ 时, } 2\pi r h D_1 = \lambda_0 h$$

$$D_1 = \frac{\lambda_0}{2\pi r} \quad E_1 = \frac{D_1}{\epsilon_0 \epsilon_{r1}} = \frac{\lambda_0}{2\pi \epsilon_0 \epsilon_{r1} r}$$

$$R_1 < r < R_2 \text{ 时, } 2\pi r h D_2 = \lambda_0 h \quad D_2 = \frac{\lambda_0}{2\pi r}$$

$$E_2 = \frac{D_2}{\epsilon_0 \epsilon_{r2}} = \frac{\lambda_0}{2\pi \epsilon_0 \epsilon_{r2} r}$$

(2) 导线和圆筒间的电势差

$$\begin{aligned} U &= \int_{(R_0)}^{(R_2)} \vec{E} \cdot d\vec{l} = \int_{(R_0)}^{(R_1)} \vec{E}_1 \cdot d\vec{l} + \int_{(R_1)}^{(R_2)} \vec{E}_2 \cdot d\vec{l} = \int_{R_0}^{R_1} \frac{\lambda_0}{2\pi\epsilon_0\epsilon_{r1}r} dr + \int_{R_1}^{R_2} \frac{\lambda_0}{2\pi\epsilon_0\epsilon_{r2}r} dr \\ &= \frac{\lambda_0}{2\pi\epsilon_0\epsilon_{r1}} \ln \frac{R_1}{R_0} + \frac{\lambda_0}{2\pi\epsilon_0\epsilon_{r2}} \ln \frac{R_2}{R_1} = \frac{\lambda_0}{2\pi\epsilon_0\epsilon_{r1}\epsilon_{r2}} \left(\epsilon_{r2} \ln \frac{R_1}{R_0} + \epsilon_{r1} \ln \frac{R_2}{R_1} \right) \end{aligned}$$

(3) 介质内的 $\mathbf{P}$ 及介质表面的极化电荷面密度 σ'

$$P_1 = \epsilon_0(\epsilon_{r1} - 1)E_1 = \epsilon_0(\epsilon_{r1} - 1) \frac{\lambda_0}{2\pi\epsilon_0\epsilon_{r1}r} = \frac{(\epsilon_{r1} - 1)\lambda_0}{2\pi\epsilon_{r1}r}$$

$$P_2 = \epsilon_0(\epsilon_{r2} - 1)E_2 = \epsilon_0(\epsilon_{r2} - 1) \frac{\lambda_0}{2\pi\epsilon_0\epsilon_{r2}r} = \frac{(\epsilon_{r2} - 1)\lambda_0}{2\pi\epsilon_{r2}r}$$

$$\sigma'_1 = P_1(R_0) \cos \pi = -\frac{(\epsilon_{r1} - 1)\lambda_0}{2\pi\epsilon_{r1}R_0} \quad \sigma'_3 = P_2(R_2) \cos 0 = \frac{(\epsilon_{r2} - 1)\lambda_0}{2\pi\epsilon_{r2}R_2}$$

$$\sigma'_2 = P_1(R_1) \cos 0 + P_2(R_1) \cos \pi = \frac{(\epsilon_{r1} - 1)\lambda_0}{2\pi\epsilon_{r1}R_1} - \frac{(\epsilon_{r2} - 1)\lambda_0}{2\pi\epsilon_{r2}R_1} = \frac{(\epsilon_{r1} - \epsilon_{r2})\lambda_0}{2\pi\epsilon_{r1}\epsilon_{r2}R_1}$$

(4) 该电容器的电容C

$$\begin{aligned} C &= \frac{\lambda_0 l}{U} = \lambda_0 l \cdot \left[\frac{\lambda_0}{2\pi\epsilon_0\epsilon_{r1}\epsilon_{r2}} \left(\epsilon_{r2} \ln \frac{R_1}{R_0} + \epsilon_{r1} \ln \frac{R_2}{R_1} \right) \right]^{-1} \\ &= \frac{2\pi\epsilon_0\epsilon_{r1}\epsilon_{r2}l}{\epsilon_{r2} \ln(R_1/R_0) + \epsilon_{r1} \ln(R_2/R_1)} \end{aligned}$$

B1-1. 如图所示, 三个“无限长”的同轴导体圆柱面A,B和C, 半径分别为 R_a , R_b 和 R_c , 圆柱面B上带电荷, A和C都接地. 同轴导体圆柱面之间为真空, 求B的内表面上电荷线密度 λ_1 和外表面上电荷线密度 λ_2 之比值 λ_1/λ_2 .

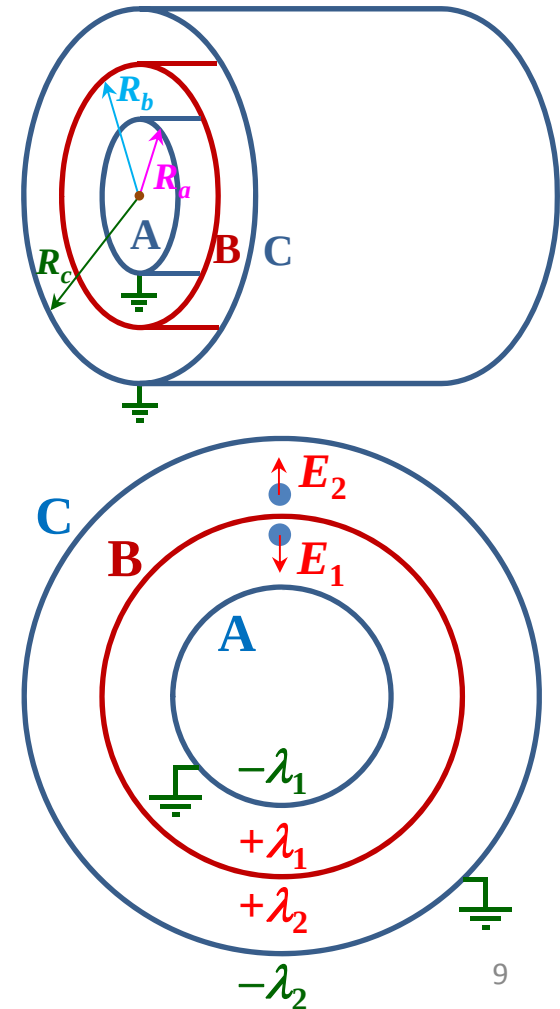
解: **解法一** 设圆柱面B上带正电荷, 其内表面上电荷线密度 λ_1 , 外表面上电荷线密度 λ_2 , 由于A和C都接地, 如图所示,

则A和B之间的电场强度为

$$E_1 = \lambda_1 / 2\pi\epsilon_0 r, \quad \text{则方向由B指向A}$$

则B和C之间的电场强度为

$$E_2 = \lambda_2 / 2\pi\epsilon_0 r, \quad \text{则方向由B指向C}$$



则**B**和**A**之间的电势差为

$$U_{BA} = \int_{R_b}^{R_a} \vec{E}_1 \cdot d\vec{l} = \int_{R_b}^{R_a} \frac{\lambda_1}{2\pi\epsilon_0 r} \cos \pi dr$$

$$= -\frac{\lambda_1}{2\pi\epsilon_0} \int_{R_b}^{R_a} \frac{1}{r} dr = \frac{\lambda_1}{2\pi\epsilon_0} \ln \frac{R_b}{R_a}$$

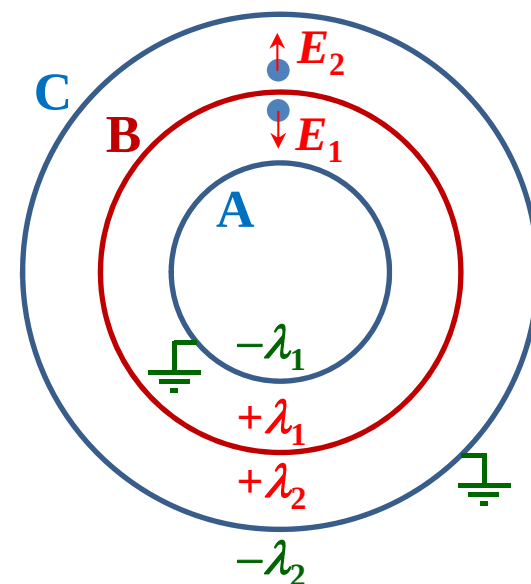
则**B**和**C**之间的电势差为

$$U_{BC} = \int_{R_b}^{R_c} \vec{E}_2 \cdot d\vec{l} = \int_{R_b}^{R_c} \frac{\lambda_2}{2\pi\epsilon_0 r} \cos 0^\circ dr$$

$$= \frac{\lambda_2}{2\pi\epsilon_0} \int_{R_b}^{R_c} \frac{1}{r} dr = \frac{\lambda_2}{2\pi\epsilon_0} \ln \frac{R_c}{R_a}$$

因 $U_{BA} = U_{BC}$, 得到

$$\frac{\lambda_1}{\lambda_2} = \frac{\ln(R_c/R_b)}{\ln(R_b/R_a)}$$



解法二 设长为 l 的一段导体,

电容BA的带电量为

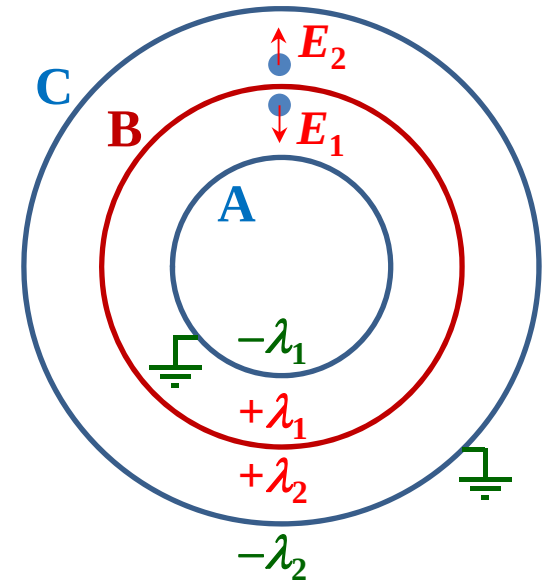
$$Q_{BA} = C_{BA} U_{BA}$$

电容BC的带电量为

$$Q_{BC} = C_{BC} U_{BC}$$

因 $U_{BA} = U_{BC}$, 得到

$$\frac{Q_{BA}}{Q_{BC}} = \frac{C_{BA}}{C_{BC}} = \frac{\lambda_1}{\lambda_2}$$



$$C_{BA} = \frac{2\pi\epsilon_0 l}{\ln R_b/R_a}$$

$$C_{BC} = \frac{2\pi\epsilon_0 l}{\ln R_c/R_b}$$

$$\frac{\lambda_1}{\lambda_2} = \frac{\ln(R_c/R_b)}{\ln(R_b/R_a)}$$

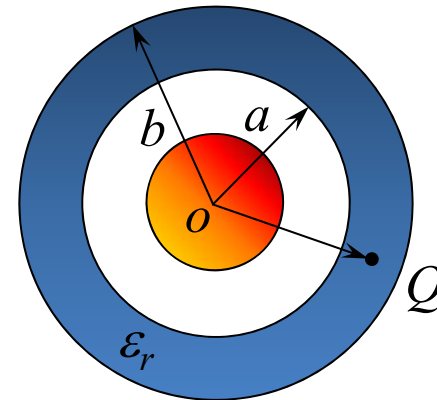
B1-2 (本题25分) 5246

设有一带有电荷 q (>0)的金属球被内外半径分别为 a 和 b 的同心电介质球壳所包围 (如图所示), 电介质的相对介电常数为 ϵ_r . 求: (1) 电介质球壳中任一点 Q 的电位移 D 、电场强度 E 、电势 U 及极化强度 P 与 Q 点到球心 o 的距离 r 间的关系; (2) 电介质球壳内外表面上的极化电荷面密度.

解:
$$\oiint_S \vec{D} \cdot d\vec{S} = D \cdot 4\pi r^2 = q$$

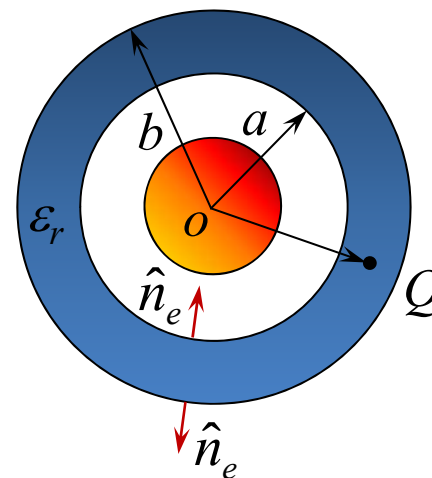
$$D = \frac{q}{4\pi r^2}$$

方向向外



$$E = \frac{D}{\varepsilon_0 \varepsilon_r} = \frac{q}{4\pi \varepsilon_0 \varepsilon_r r^2} \quad \text{方向向外}$$

$$P = \varepsilon_0 (\varepsilon_r - 1) E = \frac{q}{4\pi r^2} \left(1 - \frac{1}{\varepsilon_r}\right) \quad \text{方向向外}$$



$$\sigma'_{\text{内}} = P \cos \pi \Big|_{r=a} = -P \Big|_{r=a} = -\frac{q}{4\pi a^2} \left(1 - \frac{1}{\varepsilon_r}\right)$$

$$\sigma'_{\text{外}} = P \cos 0 \Big|_{r=b} = P \Big|_{r=b} = \frac{q}{4\pi b^2} \left(1 - \frac{1}{\varepsilon_r}\right)$$

$$E_4 = \frac{D}{\varepsilon_0} = \frac{q}{4\pi\varepsilon_0 r^2}$$

$$U_3 = \int_r^\infty \vec{E} \bullet d\vec{r} = \int_r^b E dr + \int_b^\infty E_4 dr$$

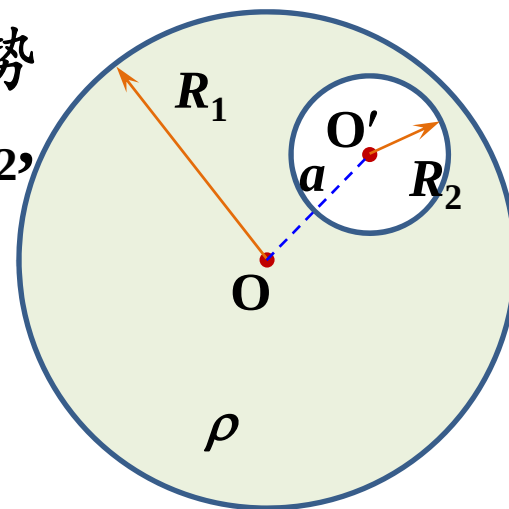
$$= \int_r^b \frac{q}{4\pi\varepsilon_0 \varepsilon_r r^2} dr + \int_b^\infty \frac{q}{4\pi\varepsilon_0 r^2} dr$$

$$= \frac{q}{4\pi\varepsilon_0 \varepsilon_r} \left(\frac{1}{r} - \frac{1}{b} \right) + \frac{q}{4\pi\varepsilon_0} \left(\frac{1}{b} - 0 \right)$$

$$= \frac{q}{4\pi\varepsilon_0 \varepsilon_r} \left(\frac{1}{r} + \frac{\varepsilon_r - 1}{b} \right)$$

C1-1. 如图所示,一半径为 R_1 的球体均匀带电,电荷体密度为 ρ ,球内有一半径为 R_2 的球形空腔,空腔中心 O' 与球心 O 相距为 a ,若无穷远处取为电势零点,试求空腔中心点 O' 处的电势。

解: 由补偿法分析,球形空腔中场点 P 的电势是半径为 R_1 ,密度为 ρ 的大球和半径为 R_2 ,密度为 $-\rho$ 的小球产生的电势之和,即



方法一:

大球的电场分布为

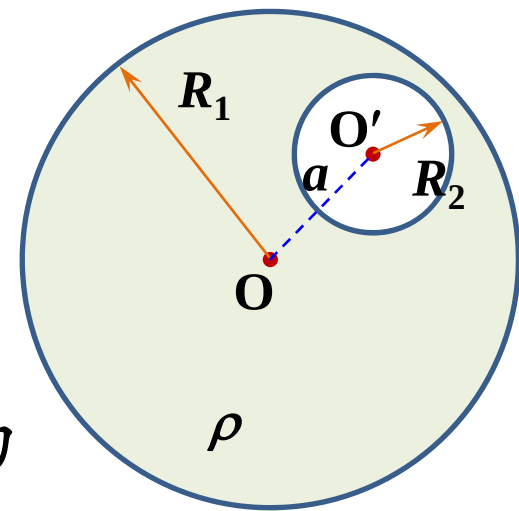
$$\vec{E} = \begin{cases} \frac{\rho}{3\epsilon_0} \vec{r} & r < R_1 \\ \frac{\rho R_1^3}{3\epsilon_0 r^3} \vec{r} & r > R_1 \end{cases} \quad \begin{array}{l} \text{应用电势定义,可求大球内任一点} \\ \text{的电势为} \end{array}$$

$$U_1 = \int_r^{R_1} \vec{E}_1 \cdot d\vec{r} + \int_{R_1}^{\infty} \vec{E}_1 \cdot d\vec{r} = \frac{\rho}{6\epsilon_0} (3R_1^2 - r^2)$$

大球在空腔中心点 O' ($r=a$)处的电势

$$U_{1O'} = \frac{\rho}{6\epsilon_0} (3R_1^2 - a^2)$$

小球的电场分布为 $\vec{E}_2 = \begin{cases} \frac{-\rho}{3\varepsilon_0} \vec{r}' & r' < R_2 \\ \frac{-\rho R_2^3}{3\varepsilon_0 r'^3} \vec{r}' & r' > R_2 \end{cases}$



应用电势定义, 可求小球内任一点的电势为

$$U_2 = \int_{r'}^{R_2} \vec{E}_2 \cdot d\vec{r} + \int_{R_2}^{\infty} \vec{E}_2 \cdot d\vec{r} = \frac{-\rho}{6\varepsilon_0} (3R_2^2 - r'^2)$$

小球在空腔中心点 O' ($r' = 0$) 处的电势

$$U_{2O'} = \frac{-\rho}{6\varepsilon_0} 3R_2^2 = \frac{-\rho R_2^2}{2\varepsilon_0}$$

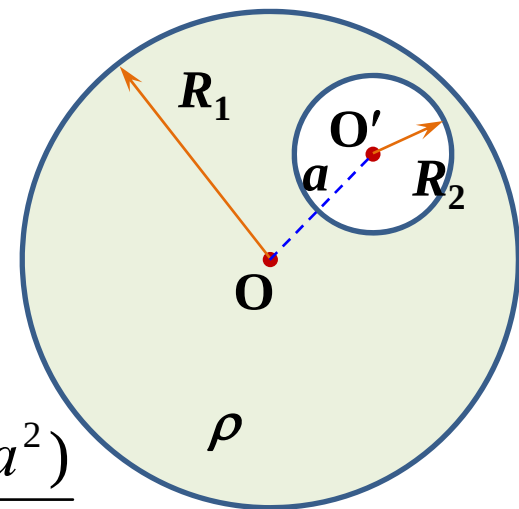
由电势叠加原理

$$U_{O'} = U_{1O'} + U_{2O'} = \frac{\rho}{6\varepsilon_0} (3R_1^2 - a^2) + \left(\frac{-\rho R_2^2}{2\varepsilon_0} \right) = \frac{\rho}{6\varepsilon_0} [3R_1^2 - 3R_2^2 - a^2]$$

方法二:

由电势叠加原理大球在空腔
中心点 O' ($r=a$)处的电势

$$\begin{aligned} U_{1O'} &= \int_0^{(a)} \frac{dq}{4\pi\epsilon_0 a} + \int_{(a)}^{(R_1)} \frac{dq}{4\pi\epsilon_0 r} \\ &= \int_0^a \frac{\rho 4\pi r^2 dr}{4\pi\epsilon_0 a} + \int_a^{R_1} \frac{\rho 4\pi r^2 dr}{4\pi\epsilon_0 r} = \frac{\rho a^2}{3\epsilon_0} + \frac{\rho(R_1^2 - a^2)}{2\epsilon_0} \\ &= \frac{\rho}{6\epsilon_0} (3R_1^2 - a^2) \end{aligned}$$



同理小球在空腔中心点 O' ($r'=0$)处的电势

$$U_{2O'} = \int_0^{(R_2)} \frac{dq}{4\pi\epsilon_0 r'} = \int_0^{R_2} \frac{(-\rho) 4\pi r'^2 dr'}{4\pi\epsilon_0 r'} = \frac{(-\rho) R_2^2}{2\epsilon_0}$$

由电势叠加原理

$$U_{O'} = U_{1O'} + U_{2O'} = \frac{\rho}{6\epsilon_0} (3R_1^2 - a^2) + \left(\frac{-\rho R_2^2}{2\epsilon_0} \right) = \frac{\rho}{6\epsilon_0} [3R_1^2 - 3R_2^2 - a^2]$$

C1-2. 圆柱形电容器由半径为 a 的圆柱导体及共轴的导体圆筒构成, 圆筒的内半径为 b 、圆柱导体和导体圆筒的长度均为 L , 且 $L \gg b$, 其间充满相对介电常量为 ϵ_r 的电介质. 设导体圆柱和圆筒的电势差为 U , 试求: (1) 电介质中的电场强度; (2) 电介质表面的极化电荷的面密度.

解: (1) 设圆柱形电容器内圆柱导体带电线密度为 λ , 用高斯定理可求

$$a < r < b \text{ 时, } \quad D = \frac{\lambda}{2\pi r} \quad E = \frac{D}{\epsilon_0 \epsilon_r} = \frac{\lambda}{2\pi \epsilon_0 \epsilon_r r}$$

$$U = \int_{(a)}^{(b)} \vec{E} \cdot d\vec{l} = \int_a^b \frac{\lambda}{2\pi \epsilon_0 \epsilon_r r} dr = \frac{\lambda}{2\pi \epsilon_0 \epsilon_r} \ln \frac{b}{a}$$

$$\lambda = 2\pi \epsilon_0 \epsilon_r \frac{U}{\ln(b/a)} \quad E = \frac{U}{r \ln(b/a)}$$

$$(2) \quad P = \varepsilon_0(\varepsilon_r - 1)E = \frac{\varepsilon_0(\varepsilon_r - 1)U}{r \ln(b/a)}$$

介质内表面的极化电荷的面密度

$$\sigma'_1 = P_a \cos \pi = -\frac{\varepsilon_0(\varepsilon_r - 1)U}{a \ln(b/a)}$$

介质外表面的极化电荷的面密度

$$\sigma'_2 = P_b \cos 0^\circ = \frac{\varepsilon_0(\varepsilon_r - 1)U}{b \ln(b/a)}$$